

$$1a) \quad 0 \cdot (-2) + 0.5 \cdot 0 + 1 \cdot (-1) + 1 \cdot 0 + 1 \cdot 1 + 0.5 \cdot 0 + 1 \cdot (-1) + 0.5 \cdot 0 + 0 \cdot 1 = -1$$

$$\downarrow$$

$$G_1 = \begin{bmatrix} -1 & -0.5 & -2 & -0.5 & 1 \\ -2.5 & -1 & 0 & -1.5 & -0.5 \\ -2.5 & -2 & 2 & -0.5 & -0.5 \\ 0 & -2 & -1 & -1 & -1.5 \\ 1 & 0.5 & -1 & 0 & -3 \end{bmatrix}$$

$$b) \quad X_{\text{pad}=1} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & .5 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & .5 & 0 & 0 & .5 & 1 \\ 0 & 1 & .5 & 0 & .5 & 0 & 0 & .5 \\ 0 & .5 & 0 & 0 & 1 & .5 & 0 & 1 \\ 0 & .5 & .5 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & .5 & 0 & 0 & 0 & .5 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$-2 \cdot 0 + 0 \cdot 0 - 1 \cdot 0 + 0 \cdot 0 + 1 \cdot 0 + 0 \cdot .5 - 1 \cdot 0 + 0 \cdot 1 + 1 \cdot 1 = 1$$

$$\downarrow$$

$$G = \begin{bmatrix} 1 & 0 & 0 & -0.5 & 0.5 & 1 & -0.5 \\ 1 & -1 & -0.5 & -2 & -0.5 & 1 & 1 \\ 0 & -2.5 & -1 & 0 & -1.5 & -0.5 & -0.5 \\ 0.5 & -2.5 & -2 & 2 & -0.5 & -0.5 & 1 \\ 1.5 & 0 & -2 & -1 & -1 & -1.5 & 1 \\ -0.5 & 1 & 0.5 & -1 & 0 & -3 & -0.5 \\ -1 & -0.5 & -1 & 0 & 1 & 0.5 & 1 \end{bmatrix}$$

c) G_1 : apply kernel G_2 : apply max pooling

$$0 \cdot (-2) + 0.5 \cdot 0 + 1 \cdot (-1) + 1 \cdot 0 + 1 \cdot 1 + 0.5 \cdot 0 + 1 \cdot (-1) + 0.5 \cdot 0 + 0 \cdot 1 = -1$$

$$\downarrow$$

$$G_1 = \begin{bmatrix} -1 & -2 & 1 \\ -2.5 & 2 & -0.5 \\ 1 & -1 & -3 \end{bmatrix}$$

$$\max(-1, -2, -2.5, 2) = 2$$

$$\downarrow$$

$$G_2 = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

d) G_1 is the same as G from part (b)

$$G_1 = \begin{bmatrix} 1 & 0 & 0 & -0.5 & 0.5 & 1 & -0.5 \\ 1 & -1 & -0.5 & -2 & -0.5 & 1 & 1 \\ 0 & -2.5 & -1 & 0 & -1.5 & -0.5 & -0.5 \\ 0.5 & -2.5 & -2 & 2 & -0.5 & -0.5 & 1 \\ 1.5 & 0 & -2 & -1 & -1 & -1.5 & 1 \\ -0.5 & 1 & 0.5 & -1 & 0 & -3 & -0.5 \\ -1 & -0.5 & -1 & 0 & 1 & 0.5 & 1 \end{bmatrix}$$

$$\max(1, 0, 0, 1, -1, -0.5, 0, -2.5, -1) = 1$$

$$\downarrow$$
$$G_2 = \begin{bmatrix} 1 & 0.5 & 1 \\ 1.5 & 2 & 1 \\ 1.5 & 1 & 1 \end{bmatrix}$$